



MINING DEALER BEST PRACTICE SERIES

Engine Wiring Harness Installation Aid - C175 Engines

Maintenance and Repair

Application

Component Life Management

Component Renewal (CRC)

MARC
Management

Engine Wiring Harness Installation Aid - C175 Engines	0
1. Introduction	1
2. Best Practice Description.....	1
3. Implementation Steps.....	5
4. Benefits	5
5. Resources Required	5
6. Supporting Attachments / References.....	5
7. Related Best Practices	6
8. Acknowledgements.....	6

DISCLAIMER: The information and potential benefits included in this document are based upon information provided by one or more Cat® dealers, and such dealer(s) opinion of “Best Practices”. Caterpillar makes no representation or warranty about the information contained in this document or the products referenced herein. Caterpillar welcomes additional “Best Practice” recommendations from our dealer network.

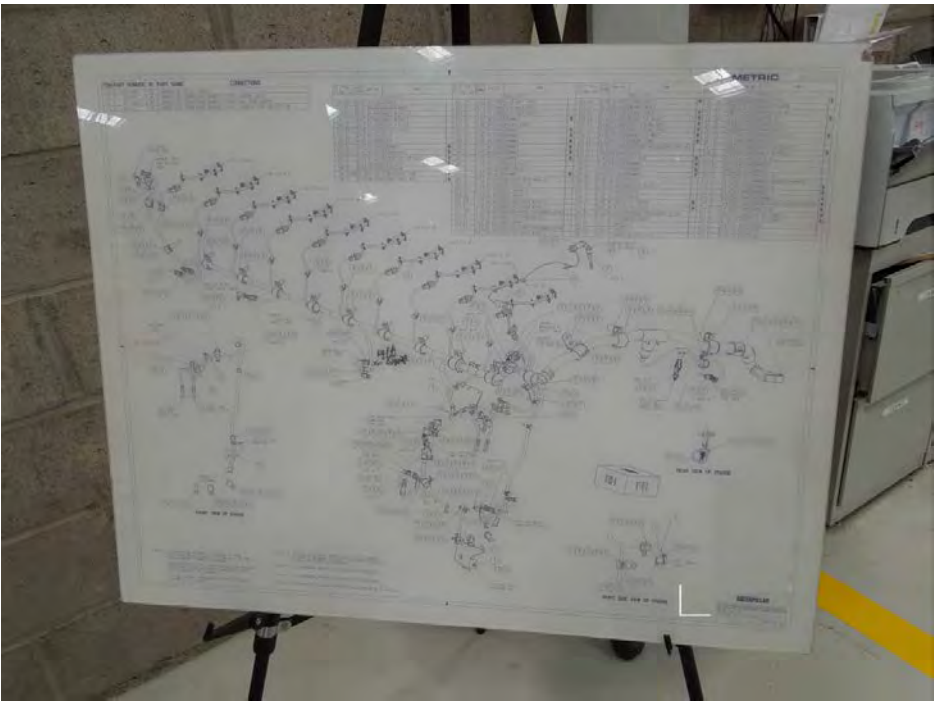
April 2016
1603-4.01-1419

1. Introduction

The engine wiring harness groups have a large number of small parts used in its installation. Matco has implemented a process to aid in the installation of the engine wiring harnesses on C175 engines. Installation of these harnesses was found to be challenging and time consuming, and a more visible guide was desired to aid the assemblers.

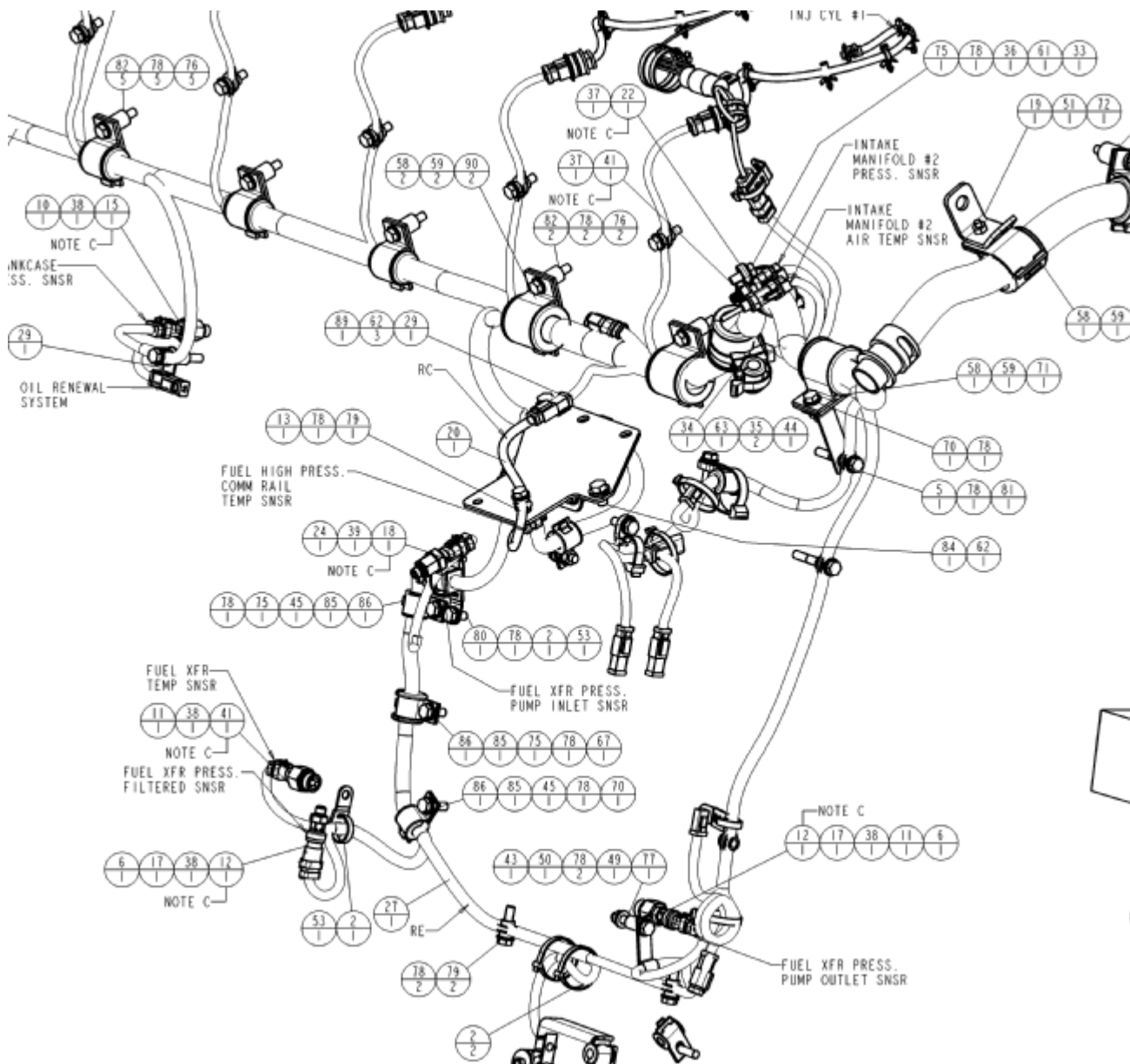
2. Best Practice Description

This Best Practice uses a print of each of the main engine wiring groups. Each print is printed on a large background using a plotter-type printer. They are then displayed temporarily in the assembly bay while the harnesses are being installed. The assembler then has a very clear, 3-D image of the harnesses and all their connectors, as well as a storage location for each part.



THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR			
Engine Wiring Harness Installation Aid - C175 Engines	DATE 4/1/2016	CHG NO 00	NUMBER 1603-4.01-1419

Close up illustrating the many small parts involved:



THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

Engine Wiring Harness Installation Aid - C175 Engines

DATE
4/1/2016

CHG
NO
00

NUMBER
1603-4.01-1419



Prior to being printed, each print is imported into Microsoft Paint program and is edited to show a cabinet location for each part. Also added is an indicator if the part is part of an assembly kit. The 'Meas Unit' field was changed to LOC (location) Cabinet. Also a line was added to create another column so that an indicator (in this case a "B") could be used to indicate that this part is included in their parts kit.

ITEM	QTY	LOC CABINET	PART NO.	NAME	
PARTS LIST					
51	5	G-6	5P-7465	CLIP-HALF TANG	
52	7	G-1	5P-7468	CLIP-HALF SLOT	
53	7	G-2	5P-7469	CLIP-HALF TANG	
54	1	C-4	5P-8112	GROMMET	B
55	4	F-19	5P-8245	WASHER-HARD	
56	1	C-2	5P-8444	GROMMET	B
57	3	C-10	5P-8939	GROMMET	B
58	4	C-13	6V-0545	GROMMET	B
59	2	C-7	6V-0683	GROMMET	B
60	6	C-9	6V-0826	GROMMET	B
61	5	C-5	6V-1249	GROMMET	B
62	1	F-10	6V-1819	BOLT M10X1.5	
63	6	F-3	6V-1820	BOLT M10X1.5	
64	1	C-6	6V-1874	GROMMET	B
				BOLT-HEX HEAD M8X1.25	

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

Engine Wiring Harness Installation Aid - C175 Engines

DATE
4/1/2016

CHG
NO
00

NUMBER
1603-4.01-1419

Cabinet drawers are organized and labelled to store replacement parts for easy retrieval and stocking.



THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

Engine Wiring Harness Installation Aid - C175 Engines

DATE
4/1/2016

CHG
NO
00

NUMBER
1603-4.01-1419

It is suggested that the prints are affixed to plexiglass for protection, and to enable simple edits with a marker as needed.



3. Implementation Steps

- Send email to Gallagher_William_L@cat.com requesting, by part number, the engine wiring group.
- Set up a cabinet of drawers with the required spare parts, divided and labelled appropriately with part number, location code, and quantity per engine.
- Open the prints in Microsoft Paint and add the location codes next to each part.
- Make large print-outs that can be referenced easily during installation of the harnesses.

4. Benefits

- Organized, easily and quickly retrieved parts.
- Clear visual representation greatly improve installation process efficiency.
- Reduced time to install C175-20 wiring harnesses from 48 man hours to 6 man hours.

5. Resources Required

- Suitable parts cabinet w/ drawers and dividers
- Resource to set up and label cabinet, and to designate cabinet locations on the print
- Process to restock as necessary

6. Supporting Attachments / References.

None

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

Engine Wiring Harness Installation Aid - C175 Engines

DATE
4/1/2016

CHG
NO
00

NUMBER
1603-4.01-1419

7. Related Best Practices

None

8. Acknowledgements

This Best Practice procedure was developed by:

Kiwi Haig, CGM

Jesus Campos, Empresas Matco S.A. de C.V

Fernando Preciado, Empresas Matco S.A. de C.V

This Best Practice was written by:

Bill Gallagher

+1 309 675 8085

[GALLAGHER WILLIAM L@cat.com](mailto:GALLAGHER_WILLIAM_L@cat.com)

Cat Global Mining

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

Engine Wiring Harness Installation Aid - C175 Engines

DATE
4/1/2016

CHG
NO
00

NUMBER
1603-4.01-1419